

ORF 445 High Frequency Trading

Homework 4

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due midnight Sunday, April 19

This homework explores two possible “signals” that can be used to forecast price motion: quote imbalance in the order book of a single asset, and a cointegration relationship between two assets that typically move together.

1. Bid-ask imbalance. As discussed in class, the “imbalance” $S = (q_B - q_A) / (q_B + q_A)$ can be used as an indicator of future price changes, especially for large-tick products.
 - (a) Pick a large-tick futures contract (see HW3). Sample the midpoint, the micro-price, and the indicator S as above on a regular grid of sample times, say at 10-second intervals from 07:00 to 15:00 on a range of dates (I recommend looping over dates in R and sending one query to Q for each date).
 - (b) For each time t , determine the direction of the *next* midpoint change after t . As discussed in class, this may be before the next sample time, or after; several sample times may use the same midpoint change. (You can do this and the previous item in one query in Q, using the `next` operator and two `aj[]`'s.)
 - (c) Group the values of the imbalance indicator S , say into 20 bins of length 0.1. Make a plot like Figure 1, showing the probability that the next midpoint change is up, conditional on the instantaneous value of the indicator being in each bin.
2. Pairs trading. Implement the algorithm described in class, and evaluate its performance using regression against forward price changes.
 - (a) For your favorite pair of assets (SOFR may work well, Treasuries a bit less well, and energies may be more challenging), extract quote midpoint prices at each of a regular set of time points. Although you will want to use the signal only during the main trading day (say 07:00 to 15:00), it is important to start the calculation at the beginning of the day so the variables can get calibrated.
 - (b) Compute the rolling averages, standard deviations, and correlation matrix for each time point throughout the day. As discussed in class, for efficiency it is important to do this by adding to the sums at each step, rather than by recomputing the sums back to the beginning on every update. For simplicity, I don't ask you to use exponential averaging or windowed averaging: at each time you use all the data back to the beginning of the day.

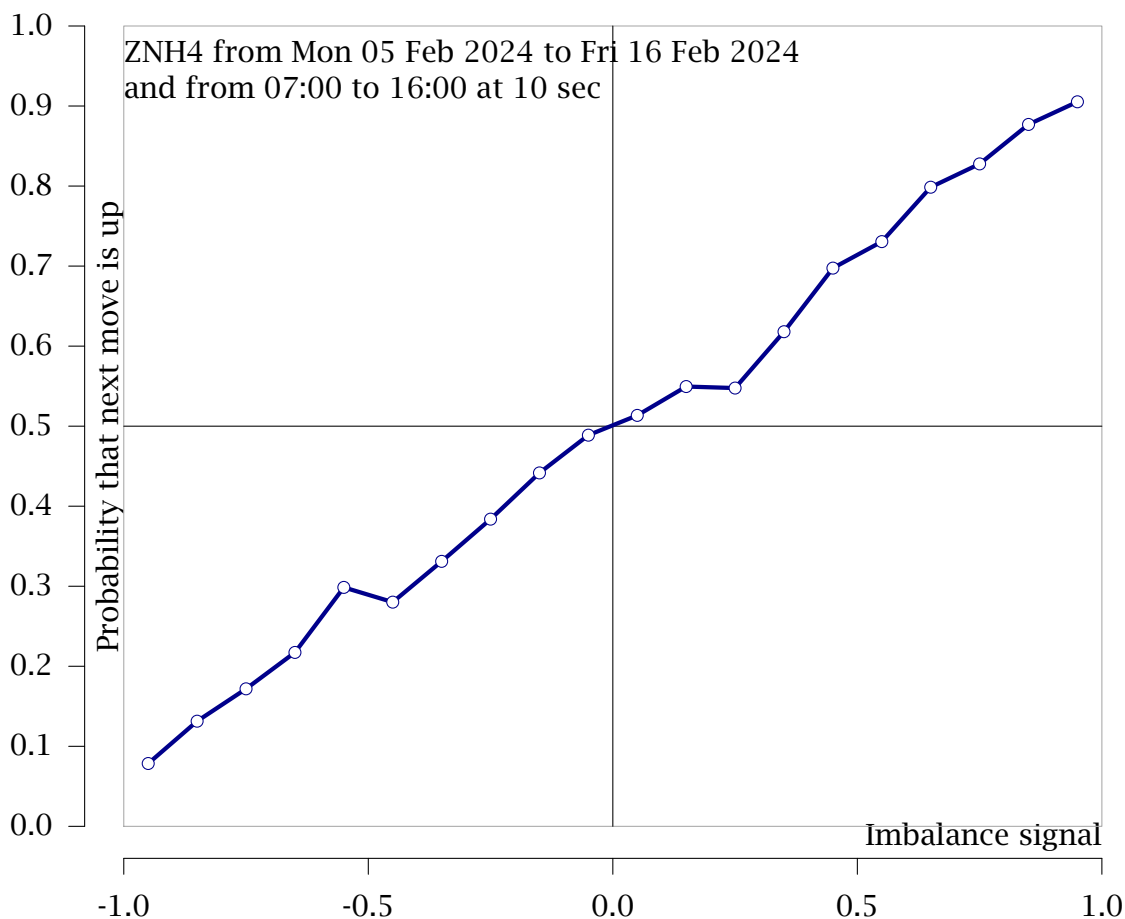


Figure 1: Probability that next midpoint change is up rather than down, as a function of the instantaneous imbalance indicator S .

- (c) Compute the SVD for the correlation matrix at each step (for a 2×2 matrix you can write it explicitly). Compute the coordinate transform as described in class. This will give you two predictors x_* and y_* for the two components of your price vector. These correspond to signals $s_x = x_* - x$ and $s_y = y_* - y$.
- (d) Regress forward price changes against the value of your signal, throughout one day. (The independent variable is your signal; the dependent variable is the forward price change on some reasonable lag such as one minute.) The results may not be as clean as the schematic pictures presented in class, but if you see anything positive that is a usable signal.